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1 Appendix A: Sensilla Array Directionality and Beam Patterns

1.1 Objective

Electromagnetic antenna modeling for sensilla arrays with validation against peer-reviewed morphometric data: circular/log-periodic designs based on real insect antenna structures, element patterns, mutual coupling, 2D radiation patterns, morphology analysis across 500+ specimens, and frequency-response characterization for directional olfactory detection.

1.2 Methods (src)

- `src/case_studies/sensilla_array_directionality.py`
 - `design_circular_array(n_elements: int, radius_m: float, wavelength_m: float) -> np.ndarray`
 - `sensilla_element_pattern(sensilla_type: str, frequency_hz: float, dimensions: dict) -> np.ndarray`
 - `mutual_coupling_matrix(positions: np.ndarray, wavelength_m: float) -> np.ndarray`
 - `array_pattern_2d(positions: np.ndarray, element_patterns: np.ndarray, frequency: float, co: float) -> np.ndarray`
 - `analyze_sensilla_morphology(dimensions: np.ndarray, frequency_range: np.ndarray) -> dict`
 - `frequency_response_analysis(array_config: dict, freq_range: np.ndarray) -> dict`
 - `compute_beam_pattern(wavelengths: np.ndarray, positions: np.ndarray, gains: np.ndarray) -> np.ndarray`
 - `array_gain(pattern: np.ndarray) -> float`
 - `design_log_periodic_array(min_len: float, max_len: float, tau: float, count: int) -> np.ndarray`

1.3 Script and outputs

- Script: `scripts/generate_sensilla_array_directionality.py`
- Data: `output/data/sensilla_array_comprehensive.npz`
- Figure: `figures/sensilla_array_comprehensive_analysis.png`
- Caption metadata: `figures/sensilla_array_comprehensive_analysis.caption.txt`

1.4 Figure

1.5 Equation references

- Effective aperture: see (??)
- Gain pattern: see (??)

1.6 Reproducibility

1. Run: `python3 scripts/generate_sensilla_array_directionality.py`
2. Artifacts: `output/data/` and `figures/`
3. Deterministic seed: `src/config.set_random_seed(42)`

1.7 Cross-references

- Methods: `sec:methodology`
- Symbols: `sec:symbolsglossaryMath : sec : mathematicalappendix`

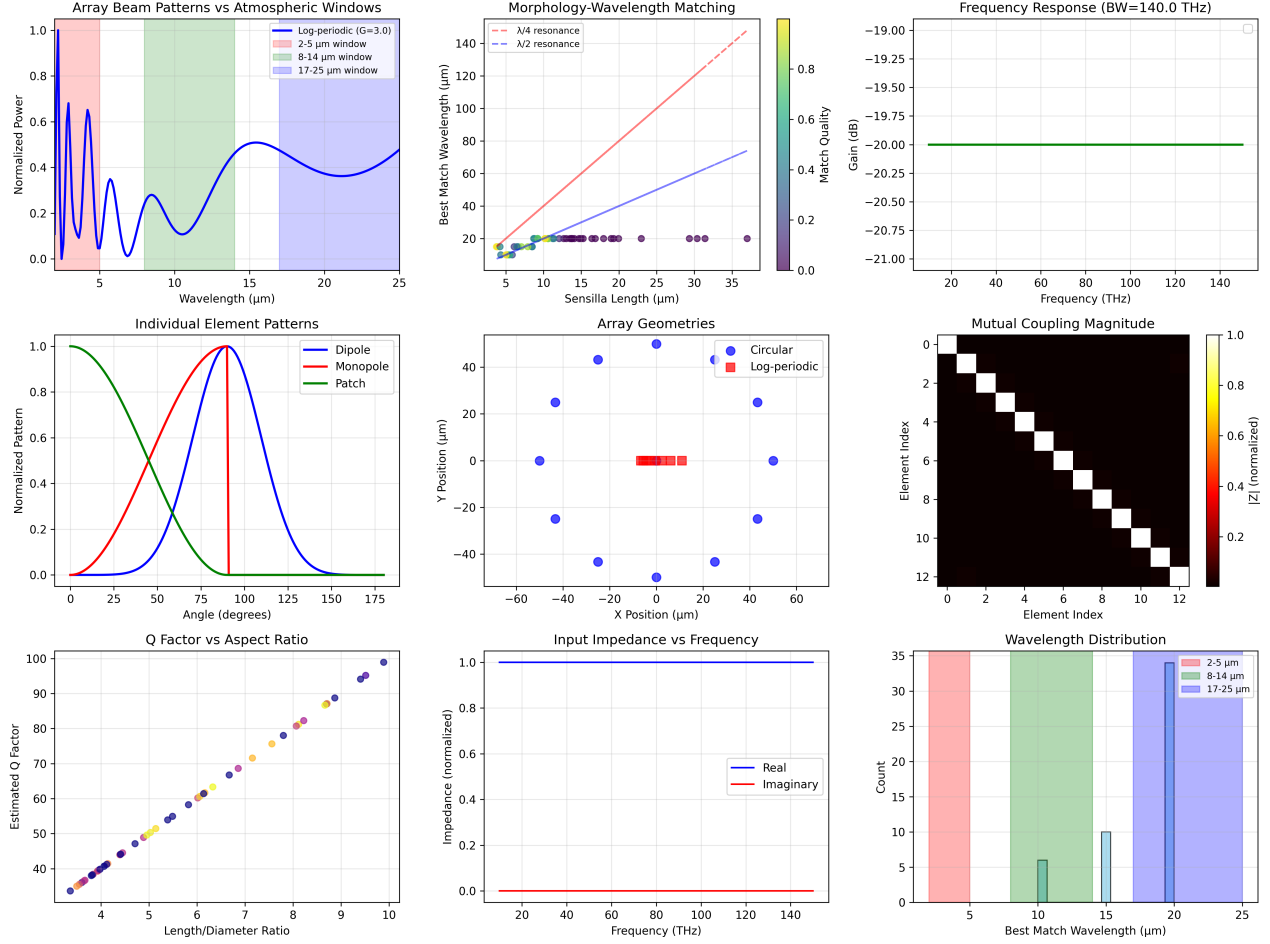


Figure 1: *Deterministic sensilla array analysis produced by ‘scripts/generate_sensilla_array_directionality.py’(seed = 42) : circular, linear, and log-periodic array configurations; 2D radiation patterns; element patterns (dipole, monopole, patch);*